

- 7 (a) Experiments were carried out to determine the effect of light intensity on the rate of photosynthesis of a species of the unicellular protist, *Chlorella*. A cell suspension of *Chlorella* was used.

Carbon dioxide uptake was used as a measure of the rate of photosynthesis.

- The suspension of *Chlorella* was illuminated at a light intensity of 3 lux for 20 seconds.
- The carbon dioxide uptake by *Chlorella* was measured at the end of the 20 second period of illumination.
- The experiment was repeated at 6 lux, 9 lux, 12 lux and in a dark room.
- The suspension was maintained at a temperature of 20 °C.

Table 7.1 shows the results of the experiments.

Table 7.1

light intensity /lux	total CO ₂ uptake after 20 seconds /μmol	rate of photosynthesis /μmol s ⁻¹
0	0	0.0
3	20	1.0
6	44	
9	72	3.6
12	80	4.0

- (i) Use Table 7.1 to calculate the rate of photosynthesis at a light intensity of 6 lux.

Complete Table 7.1 by writing your calculated value in the space provided.

[1]

- (ii) Plot a graph of the data in Table 7.1 on the grid in Fig. 7.1 to show the effect of light intensity on the rate of photosynthesis.

Draw a curve **and** extend your curve to show what would happen to the rate of photosynthesis if the experiment is carried out at 18 lux.

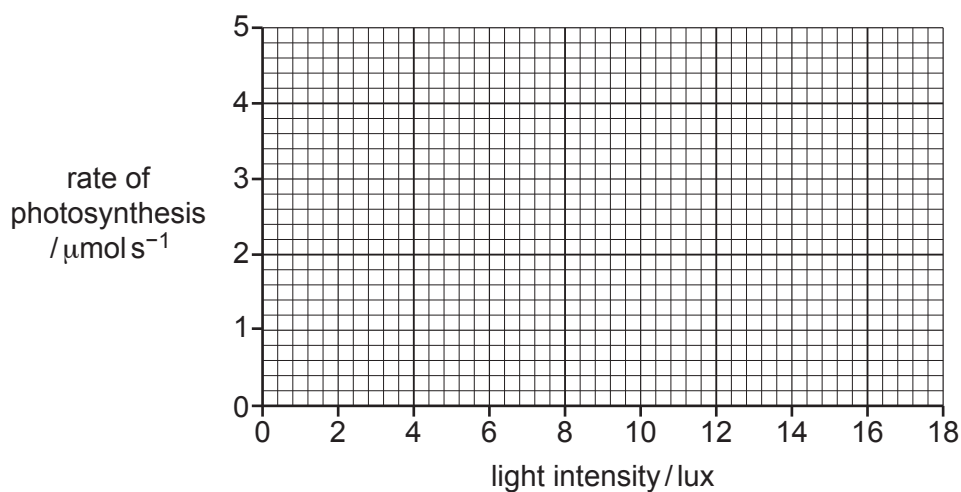


Fig. 7.1

[3]

- (iii) Suggest an explanation for the shape of your curve from 12 lux to 18 lux.

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..... [2]

(b) In photophosphorylation, photoactivation of chlorophyll results in the synthesis of ATP.

Describe how photoactivation of chlorophyll results in the synthesis of ATP in photophosphorylation.

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[Total: 10]